

GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX - DTC:
TRANSMISSION CONTROL SWITCH (TCS) [G2011723]

DESCRIPTION AND OPERATION

TRANSMISSION CONTROL SWITCH [TRANSMISSION CONTROL SWITCH (TCS)]

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- Guided Diagnostics must be followed and the repair advised by the process completed. Failure to do so may result in the rejection of any warranty claim made.
- If a control module or a component is at fault and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval program is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If Diagnostic Trouble Code(s) (DTC)s are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check Jaguar Land Rover (JLR) claims submission system for open campaigns. Refer to the corresponding bulletins and SSMS which may be valid for the specific customer complaint and complete the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be set in the Transmission Control Switch (TCS). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual. For additional information, refer to: External Controls (307-05 Automatic Transmission/Transaxle External Controls - V6 S/C 3.0L Petrol, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1142-02	Ignition Status 1 - General signal failure	- Ignition circuit short circuit to ground, short circuit to power, open circuit, high resistance - Transmission control switch internal failure	- Refer to the electrical circuit diagrams and check the ignition circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control switch
B146A-29	Control Module Integral User Switches - Signal invalid	Transmission control switch unlock switch stuck active	NOTES: - This <u>DTC</u> may be induced by the driver operating the transmission control switch unlock switch for longer than 60 seconds. - To test the operation of the transmission control switch, start the engine and move the transmission control switch to each position. - Check the operation of the transmission control switch unlock switch. Free the stuck switch or install a new transmission control switch as necessary
B146A-2A	Control Module Integral User Switches - Signal stuck in range	Transmission control switch park switch stuck active	NOTES: - This <u>DTC</u> may be induced by the driver operating the transmission control switch park switch for longer than 60 seconds. - To test the operation of the transmission control switch, start the engine and move the transmission control switch to each position. - Check the operation of the transmission control switch park switch. Free the stuck switch or install a new transmission control switch as necessary
B146A-64	Control Module Integral User Switches - Signal plausibility failure	Transmission control switch internal failure	NOTE: To test the operation of the transmission control switch, start the engine and move the transmission control switch to each position. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control switch
P062F-04	Internal Control Module EEPROM Error - System internal failures	- New transmission control switch installed and Vehicle Identification Number (VIN) not yet programmed - Transmission control switch internal failure	NOTE: To test the operation of the transmission control switch, start the engine and move the transmission control switch to each position. - Using the Jaguar Land Rover approved diagnostic equipment, configure the transmission control switch as a new module - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control switch

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P071A-02	Transmission Mode Switch A Circuit - General signal failure	Transmission control switch internal failure	NOTE: To test the operation of the transmission control switch, start the engine and move the transmission control switch to each position. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control switch
P071A-24	Transmission Mode Switch A Circuit - Signal stuck high	Transmission control switch internal failure	NOTES: - This <u>DTC</u> may be induced by the driver holding the transmission control switch in Reverse, Neutral or Drive for longer than 60 seconds. - This <u>DTC</u> may be induced by the driver holding the transmission control switch in Sport for longer than 120 seconds. - To test the operation of the transmission control switch, start the engine and move the transmission control switch to each position. - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control switch
P071A-29	Transmission Mode Switch A Circuit - Signal invalid	Transmission control switch internal failure	NOTE: To test the operation of the transmission control switch, start the engine and move the transmission control switch to each position. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control switch
P071A-48	Transmission Mode Switch A Circuit - Supervision software failure	Transmission control switch internal failure	NOTE: To test the operation of the transmission control switch, start the engine and move the transmission control switch to each position. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control switch
P071D-02	Transmission Mode Switch B Circuit - General signal failure	Transmission control switch internal failure	NOTE: To test the operation of the transmission control switch, start the engine and move the transmission control switch to each position. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control switch

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P071D-29	Transmission Mode Switch B Circuit - Signal invalid	Transmission control switch internal failure	NOTE: To test the operation of the transmission control switch, start the engine and move the transmission control switch to each position. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control switch
P0814-11	Transmission Range Display Circuit - Circuit short to ground	Transmission control switch internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform routine - On Demand Self Test. If the fault persists, install a new transmission control switch
P0814-13	Transmission Range Display Circuit - Circuit open	Transmission control switch internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform routine - On Demand Self Test. If the fault persists, install a new transmission control switch
P0814-24	Transmission Range Display Circuit - Signal stuck high	Transmission control switch internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform routine - On Demand Self Test. If the fault persists, install a new transmission control switch
P081C-64	Park Input Circuit - Signal plausibility failure	NOTE: Circuit reference - TCM P - - Park input signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - High speed Controller Area Network (CAN) bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Transmission control switch internal failure	NOTE: To test the operation of the transmission control switch, start the engine and select neutral and then select park. - Refer to the electrical circuit diagrams and check the park input signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, perform a <u>CAN</u> network integrity test. Refer to the electrical circuit diagrams and check the high speed <u>CAN</u> bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control switch
P084F-18	Park/Neutral Switch Output Circuit - Circuit current below threshold	NOTE: Circuit reference - PN SIGNAL - - Park/neutral output signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Transmission control switch internal failure	NOTE: To test the operation of the transmission control switch, start the engine and select drive and then select park. - Refer to the electrical circuit diagrams and check the park/neutral output signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control switch

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0001-82	High Speed <u>CAN</u> Communication Bus - Alive/sequence counter incorrect / not updated	Invalid data received from another control module through the high speed <u>CAN</u> bus (powertrain)	Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0001-87	High Speed <u>CAN</u> Communication Bus - Missing message	Missing message from another control module through the high speed <u>CAN</u> bus (powertrain)	Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the missing message source control module. Check the relevant control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0001-88	High Speed <u>CAN</u> Communication Bus - Bus off	High speed <u>CAN</u> bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the Jaguar Land Rover approved diagnostic equipment, perform a <u>CAN</u> network integrity test. Refer to the electrical circuit diagrams and check the high speed <u>CAN</u> bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
U0300-00	Internal Control Module Software Incompatibility - No sub type information	- Car configuration file mismatch with vehicle specification - Incorrect transmission control switch installed	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the <u>DTCs</u>. - Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the <u>DTCs</u> and retest - Check the part number of the transmission control switch. Install a new transmission control switch as necessary
U0402-00	Invalid Data Received From TCM - No sub type information	- Transmission control module power or ground circuit open circuit, high resistance - High speed <u>CAN</u> bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Transmission system fault	- Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, perform a <u>CAN</u> network integrity test. Refer to the electrical circuit diagrams and check the high speed <u>CAN</u> bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0402-2A	Invalid Data Received From TCM - Signal stuck in range	Transmission system fault	Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0402-87	Invalid Data Received From TCM - Missing message	- Transmission control module power or ground circuit open circuit, high resistance - High speed <u>CAN</u> bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Transmission system fault	- Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, perform a <u>CAN</u> network integrity test. Refer to the electrical circuit diagrams and check the high speed <u>CAN</u> bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0402-92	Invalid Data Received From TCM - Performance or incorrect operation	Transmission system fault	Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0422-00	Invalid Data Received From Body Control Module - No sub type information	Missing/invalid data from the central junction box	Using the Jaguar Land Rover approved diagnostic equipment, check the central junction box for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U2011-02	Motor - General signal failure	- Transmission control switch movement obstructed - Transmission control switch internal failure	NOTE: This <u>DTC</u> may be induced by the driver holding the transmission control switch and opposing its automatic movement. - Check the operation of the transmission control switch and make sure nothing is jamming or obstructing the mechanism. Rectify as necessary - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control switch
U2014-01	Control Module Hardware - General electrical failure	Transmission control switch internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control switch
U3003-16	Battery Voltage - Circuit voltage below threshold	- Transmission control switch power or ground circuit open circuit, high resistance - Battery/charging system fault	- Refer to the electrical circuit diagrams and check the transmission control switch power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary - Refer to the relevant section of the workshop manual and check the battery and charging system. Rectify as necessary
U3003-17	Battery Voltage - Circuit voltage above threshold	Battery/charging system fault	Refer to the relevant section of the workshop manual and check the battery and charging system. Rectify as necessary